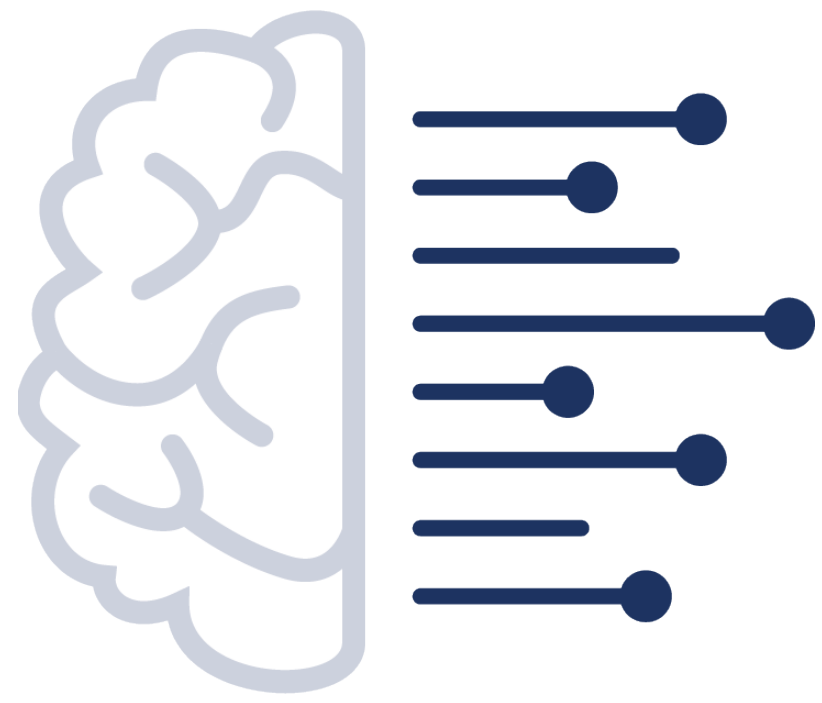


# Uncertainty Calibration of Multi-Label Bird Sound Classifiers



Raphael Schwinger<sup>1</sup>, Ben McEwen<sup>2</sup>, Vincent S. Kather<sup>2,3</sup>, René Heinrich<sup>4,5</sup>, Lukas Rauch<sup>5</sup>, and Sven Tomforde<sup>1</sup>  
<sup>1</sup>Kiel University, <sup>2</sup>Tilburg University, <sup>3</sup>Naturalis Biodiversity Center, <sup>4</sup>Fraunhofer IEE, <sup>5</sup>University of Kassel  
 raphael.schwinger@cs.uni-kiel.de

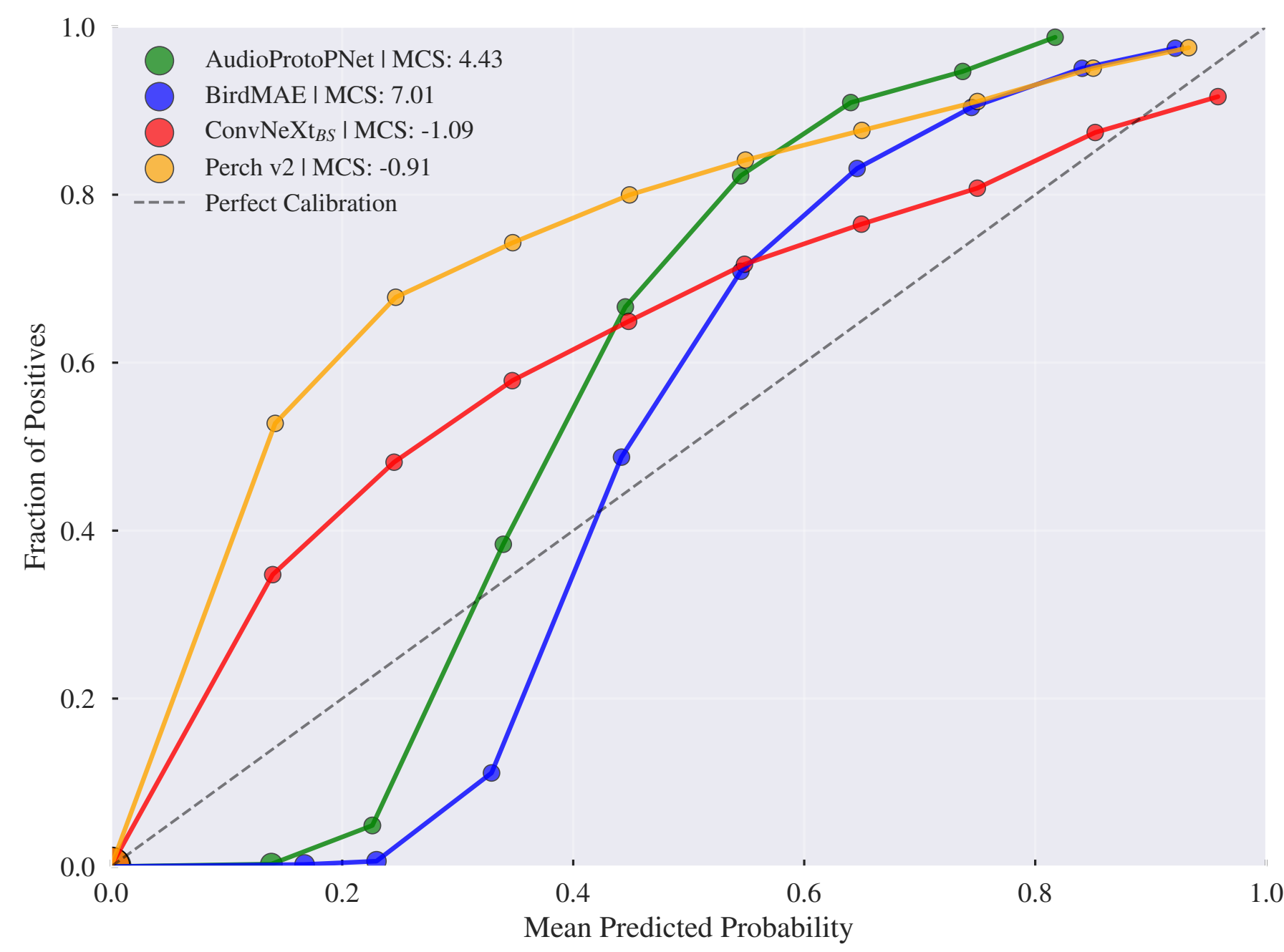
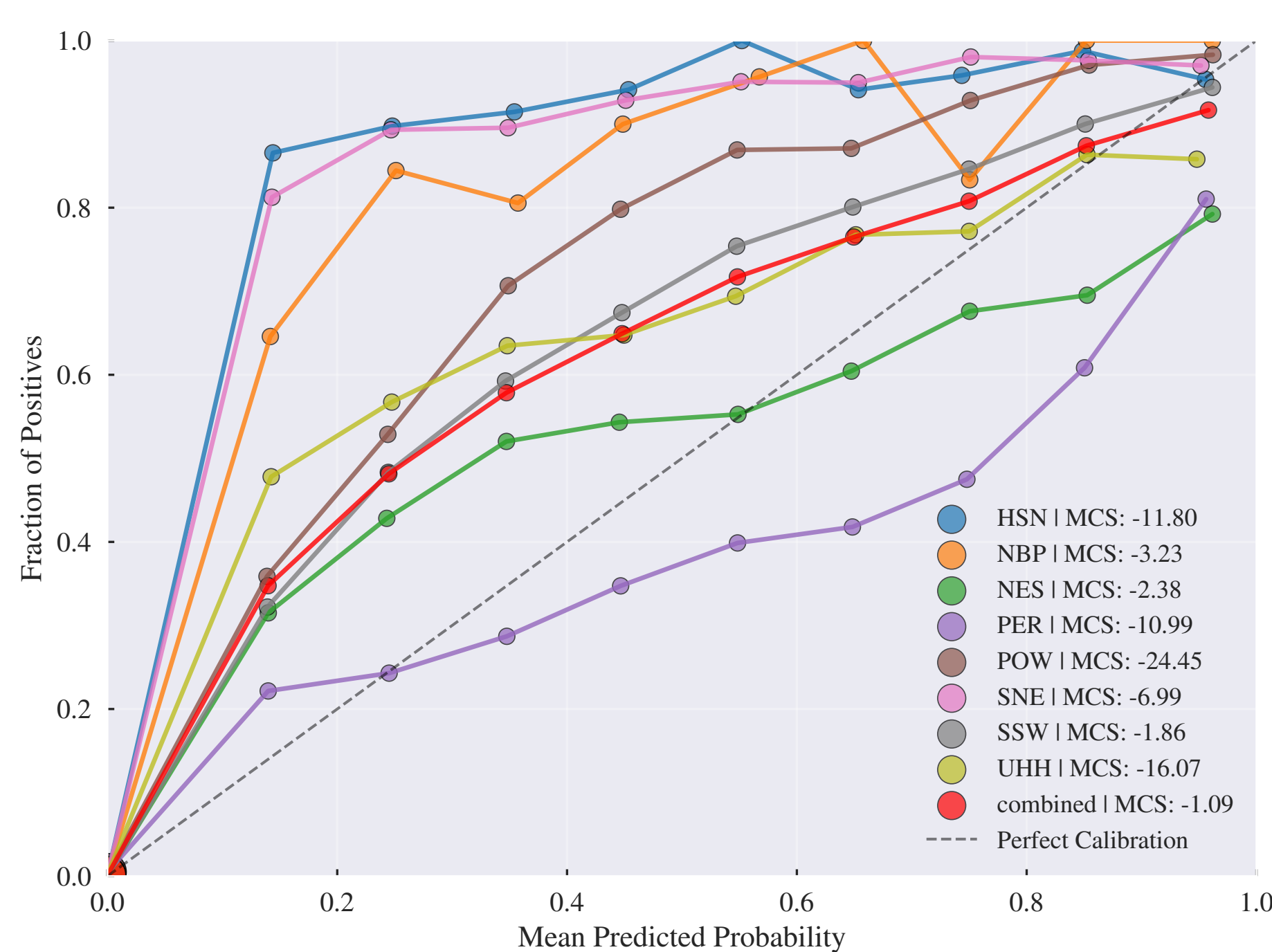


Figure 1: Reliability diagrams of bird sound classifiers on BirdSet [1].

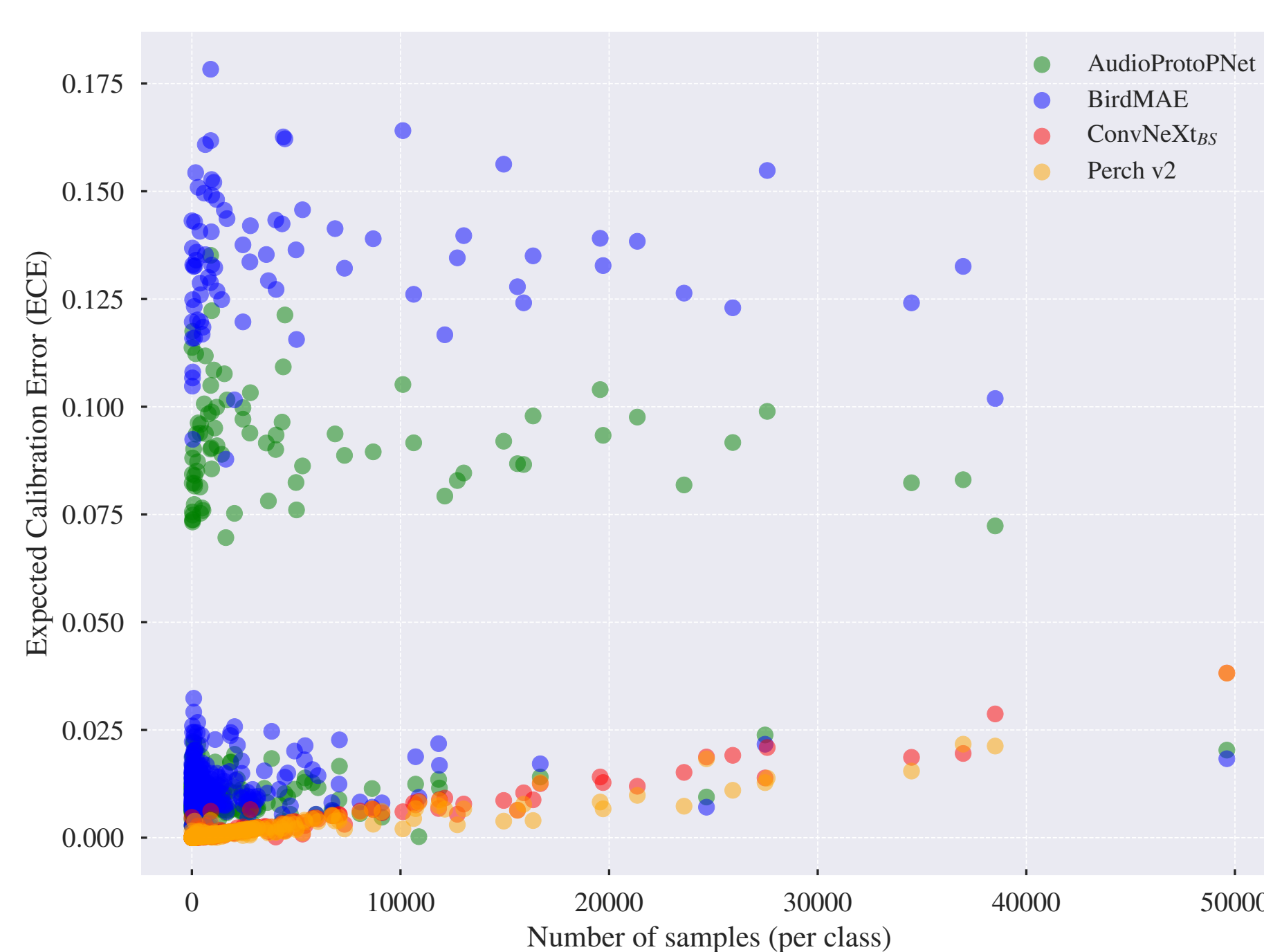
## Contributions

- **C1. Benchmark calibration** of four state-of-the-art multi-label bird sound classifiers on BirdSet [1], reporting global, per-dataset and per-class calibration with threshold-free metrics
- **C2. Investigate post hoc strategies** (temperature [5] and Platt scaling [6]): Platt scaling with 10 minutes of test data can significantly improve calibration
- **C3. Summarise implications** to give practitioners concrete suggestions for deployment-specific calibration evaluation and improvement.

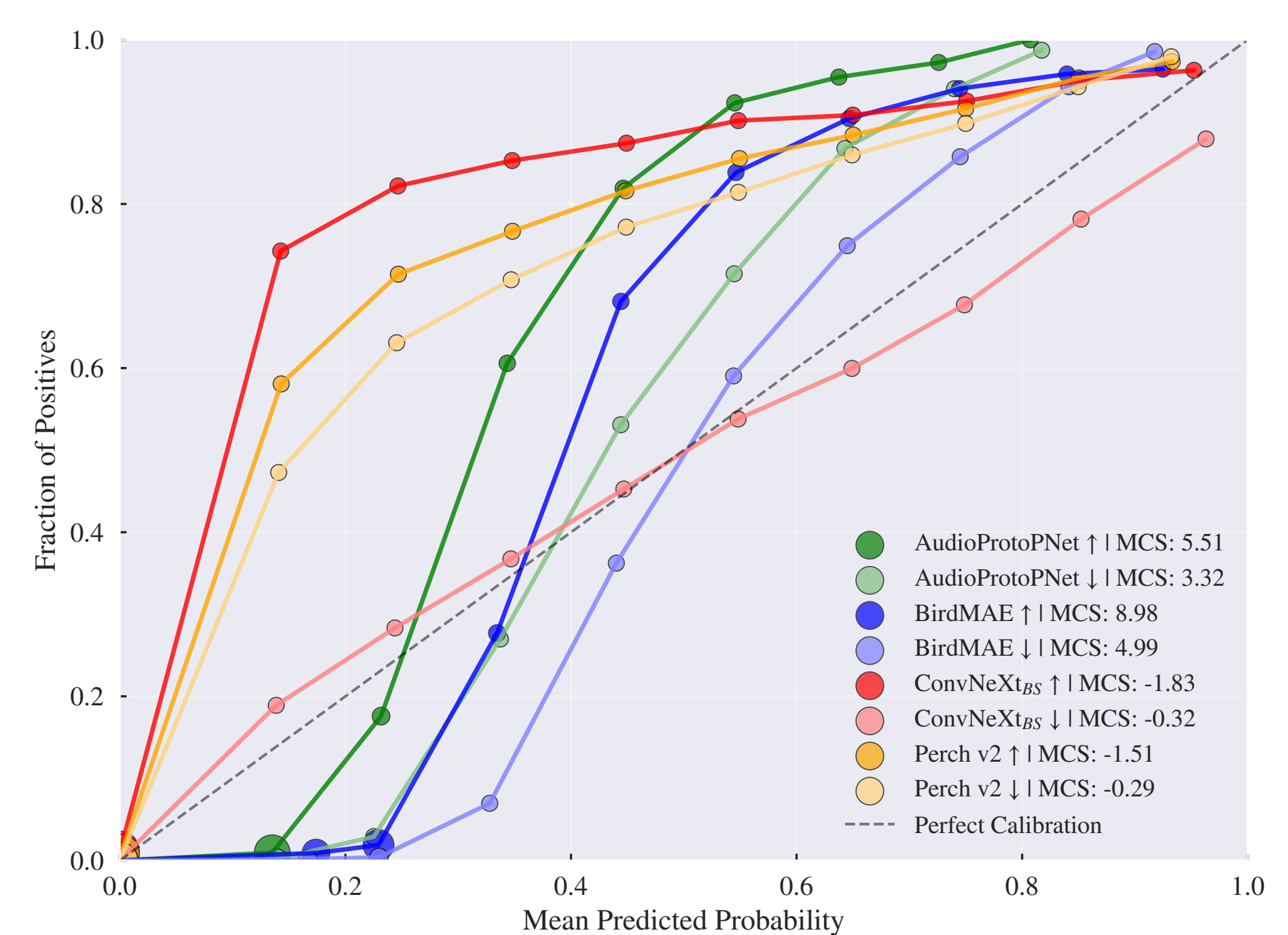
## Benchmarking the Calibration of Bird Sound Classifiers



(a) Per-dataset reliability diagram for ConvNeXtBS.



(b) Class-wise ECE vs. sample count.



(c) Reliability of most vs. least frequent classes.

### Experimental setting:

- **Models:** AudioProtoNet [3], BirdMAE [2], ConvNeXtBS [1], Perch v2 [4] (multi-label classifiers on BirdSet [1]).
- **BirdSet:** Eight expert-annotated test datasets from PAM deployments.
- **Metrics:** ECE (expected calibration error), MCS (miscalibration score), OCS/UCS (over/underconfidence); cmAP for discrimination.

### Results:

- Globally, Perch v2 and ConvNeXtBS are underconfident; AudioProtoNet and BirdMAE are overconfident (Fig. 1).
- *Calibration varies significantly per dataset* (Fig. 2a); combined ECE can mask domain-specific miscalibration.
- Class-wise ECE (Fig. 2b) shows Perch v2 and ConvNeXtBS with low ECE, increasing with the number of samples per class.
- For common vs. rare species (Fig. 2c), rare classes shift toward overconfidence for AudioProtoNet and BirdMAE; Perch v2 and ConvNeXtBS remain underconfident for both subsets.

## Post Hoc Calibration

- **Temperature scaling (TS)** and **Platt scaling (PS)** evaluated with two calibration sets: global parameters on POW and per-class parameters on the first 10 minutes of each test dataset.
- Per-class Platt scaling provides the most consistent calibration improvement across models and datasets, while even global scaling can improve calibration.

Table 1: Median relative MCS improvement (%) across test sets.

| Model         | TS POW (%) | PS POW (%) | TS/cl. (%) | PS/cl. (%) |
|---------------|------------|------------|------------|------------|
| AudioProtoNet | 56         | 56         | 75         | <b>92</b>  |
| BirdMAE       | 81         | 89         | 94         | <b>95</b>  |
| ConvNeXtBS    | 10         | 20         | <b>67</b>  | 12         |
| Perch v2      | 33         | 24         | 23         | <b>61</b>  |

## Conclusions

- **Calibration varies substantially across datasets and classes;** aggregating across domains can mask miscalibration
- **Perch v2 and ConvNeXtBS consistently underconfident;** AudioProtoNet, BirdMAE show mixed over-/underconfidence
- **Simple post hoc methods** (Platt scaling with 10 min labelled data) can **significantly improve deployment-specific calibration**
- Recommend reporting OCS/UCS alongside ECE or MCS to reveal whether errors stem from over- or underconfidence
- **Future work:** Bayesian approaches, calibration@k metrics, and evaluation across additional deployment scenarios

### KEY REFERENCES

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### MORE INFORMATION



[github.com/DBD-research-group/UncertainBird](https://github.com/DBD-research-group/UncertainBird)